

DENNIS KACHILA

BIOMEDICAL/ELECTRICAL ENGINEER

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SUMMARY

I am a Biomedical Engineer with strong Electrical and Electronics Engineering skills, supported by hands-on experience in electrical systems maintenance, control circuits, embedded systems, and CAD-based design. I have a solid background in electrical installations, troubleshooting, and preventive maintenance across hospital and industrial environments. My experience includes circuit design, single- and three-phase motor control systems, power systems interfacing, and additive manufacturing of functional engineering components. I am proficient in CAD modeling using SolidWorks and Fusion 360, embedded systems development, and technical documentation, with a strong analytical and problem-solving approach. I am adaptable, detail-oriented, and effective both as an independent engineer and as part of multidisciplinary engineering teams.

WORK EXPERIENCE

Electrical/ Electronics Engineer, Sasaflex Company Ltd

July, 2025 - Present

- Performed systems troubleshooting, electrical installation, and fault diagnosis across electrical and electronic systems.
- Conducted electronics and electrical repairs, including advanced motherboard troubleshooting and restoration of non-functional (dead) boards.
- Diagnosed circuit-level faults and carried out component-level repairs on electronic boards.
- Repaired and serviced mobile phones, laptops, and computer hardware.
- Maintained and repaired home appliances and general electrical equipment.
- Provided technical support in both workshop and field-based environments.

Lead Biomedical / Systems Engineer, Afyafab Technologies company Ltd

April 2024 - July 2025

- Led CAD-based design and prototyping of electrical and electromechanical components using SolidWorks and Fusion 360.
- Applied electrical and electronics engineering principles in system diagnostics, component selection, and functional testing.
- Designed and prototyped custom housings, fixtures, and assemblies for electrically powered systems.
- Delivered technical training and consultancy on CAD design, rapid prototyping, and basic electrical integration.

Biomedical intern, Mathari National Teaching & Referral Hospital

Jun 2022 - Aug 2022

- Performed electrical maintenance and troubleshooting on medical equipment and infrastructure systems.
- Executed preventive and corrective maintenance on motors, power supplies, and control units.
- Carried out electrical wiring layouts, appliance repair, and system inspections in compliance with safety standards.
- Supported procurement and replacement of medical spare parts.
- Applied skills in motor repair, welding, metal fabrication, and electrical installations.

Biomedical intern, The Nairobi Hospital**Jun 2022 - Aug 2022**

- Proficient in planned preventive maintenance, troubleshooting faults in medical devices, and repairing medical equipment.
- Experienced in handling job cards, conducting pre-operational checks, and refilling oxygen tanks.
- Skilled in repairing medical machines, calibrating machines, plumbing, and steam boiler management.

Engineering intern, Kenyatta University**May 2020 - Sept 2020**

- Worked with single-phase and three-phase electrical systems, wiring diagrams, and motor control circuits.
- Designed and prototyped electrical and electronic circuits, including control systems using microcontrollers.
- Developed embedded systems and IoT-based motor control solutions.
- Applied CAD tools for electrical layouts, component design, and system prototyping.
- Gained experience in power systems, starters, protection circuits, and system troubleshooting.

EDUCATION

Bachelor of Science in Biomedical Engineering**Sept 2018 - Jan 2024**

Kenyatta University

- Application of electrical engineering principles to medicine to design innovative medical devices and improve healthcare delivery systems.

Kenya Certification of Secondary Education**Feb 2014 - Nov 2017**

Kaptama Boys High School

Kenya Certification of Primary Education**Jan 2005 - Oct 2013**

Walanga Pri. School

SKILLS

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|---|---------------------------------------|--|
| • Electrical Systems Installation & Maintenance | • Embedded Systems & Microcontrollers | • Electrical Fabrication & Prototyping |
| • Circuit Design & Troubleshooting | • Control Circuits & Power Systems | • Preventive Maintenance & Fault Diagnosis |
| • Motor Control (Single & Three Phase) | • CAD Design (SolidWorks, Fusion 360) | • Technical Documentation & Reporting |

LANGUAGES

- English
- Kiswahili

REFERENCES

Ouda Jesse

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